LESSON 3 • TERM 4 • LEVEL 1

Copy on the Grid

A grid helps us say exactly where things are

TODAY WE WILL

Copy a figure exactly.

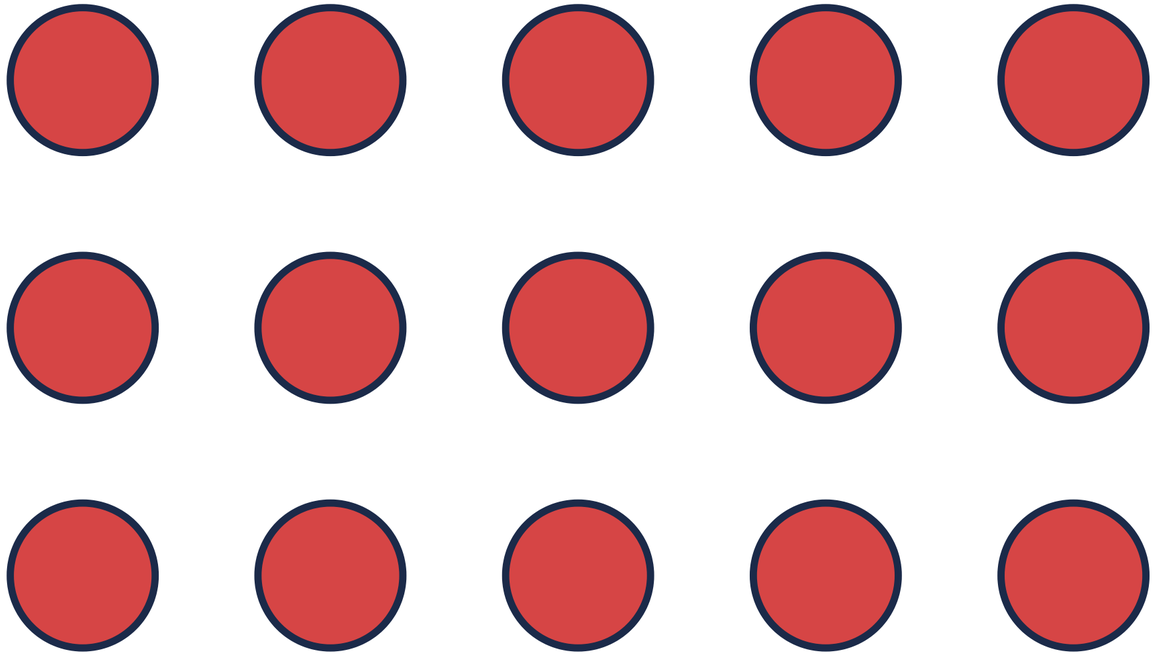
Say where to go on a grid.

Count the different routes.



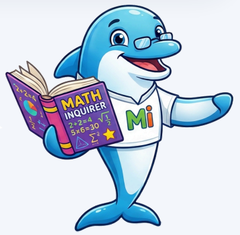
Warm-up: rows and columns

2–8 MIN



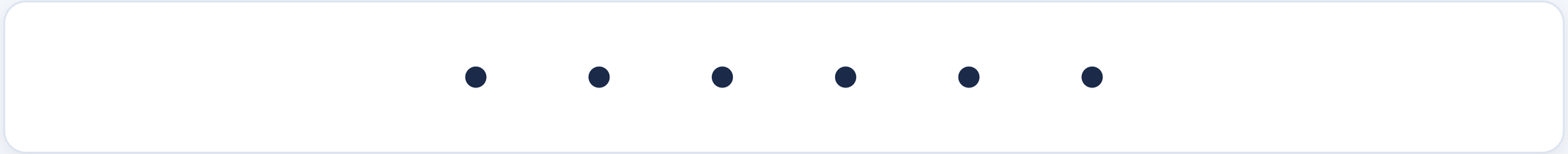
Rows go across. Columns go down. How many rows? How many in each row?

Take your time!



Steps between dots

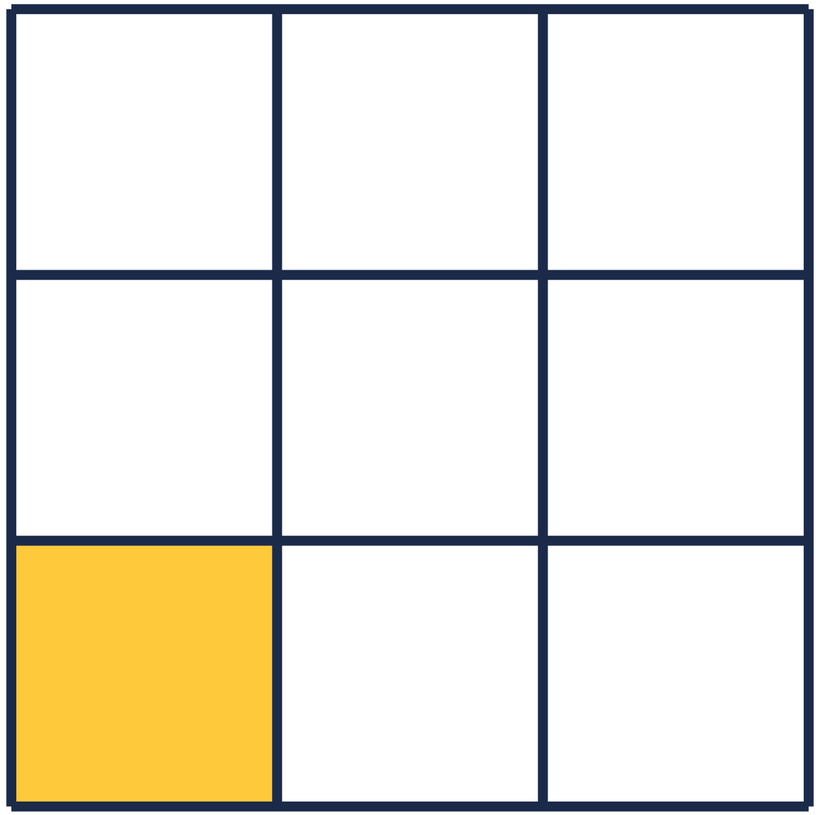
2–8 MIN



A step is the jump from one dot to the next. How many steps from the first dot to the last dot?

Right and up

2–8 MIN



The counter is in the yellow square. Move it 1 right, then 1 up. Which square?

Sneak-peek puzzles

8–12 MIN

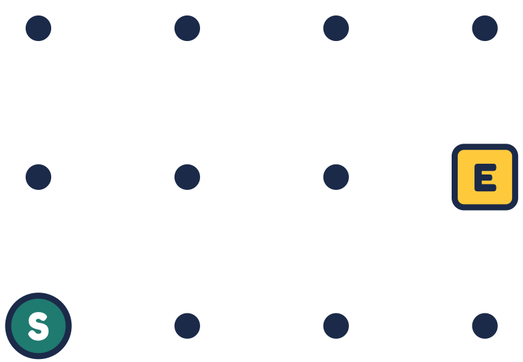
Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

The counter is on S.

It must reach E.

Write: ___ right, ___ up.

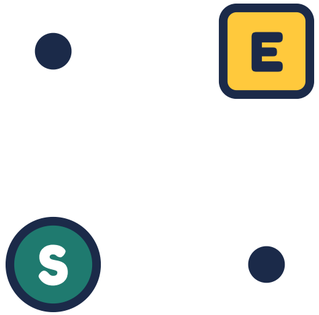


PUZZLE 2: think

The counter goes from S to E.

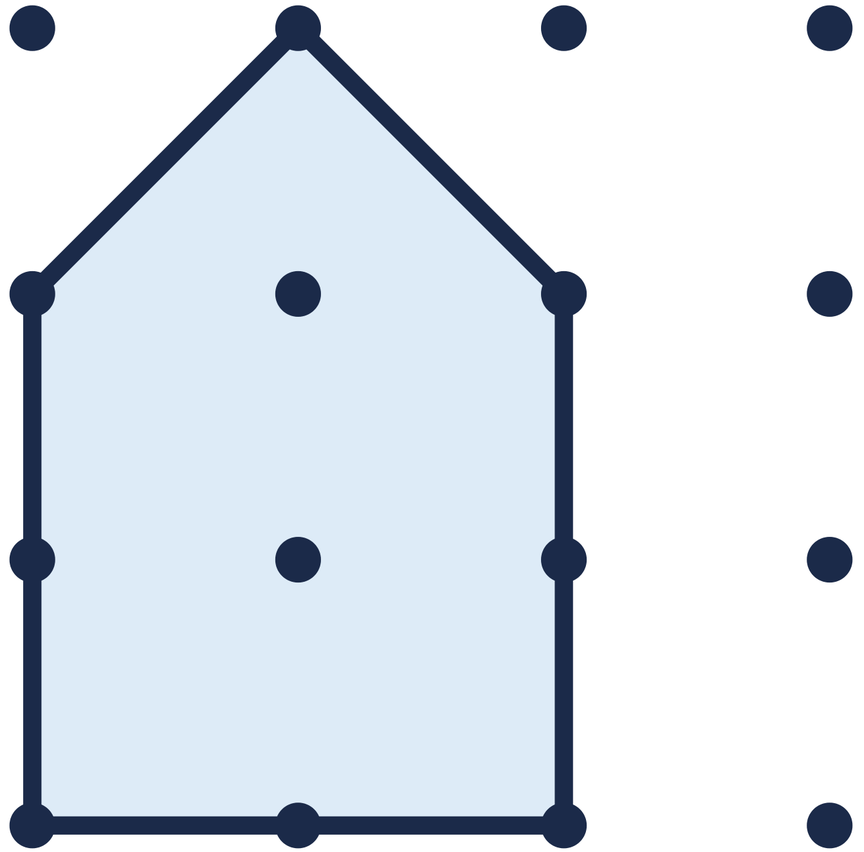
It moves only right or up.

How many different routes? I know because ___.



Explore: copy it exactly

12–23 MIN

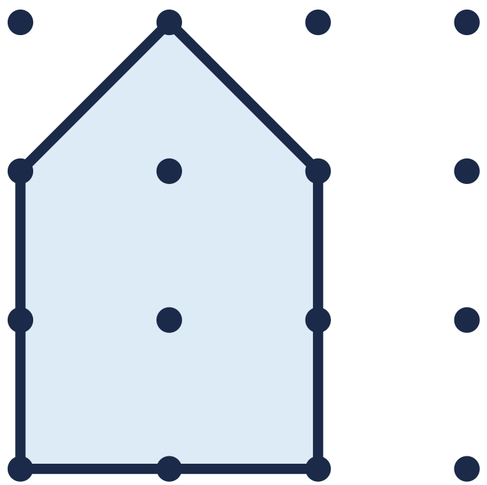


Math Inquirers Term 4 • Notice it. Test it. Explain why.

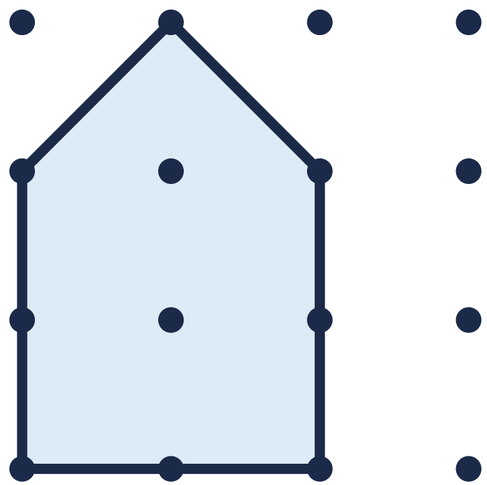
Copy the house on your copy grid. Make it exactly the same.

A copy and the figure

12–23 MIN



The figure



An exact copy

Explore: walk your button

12–23 MIN

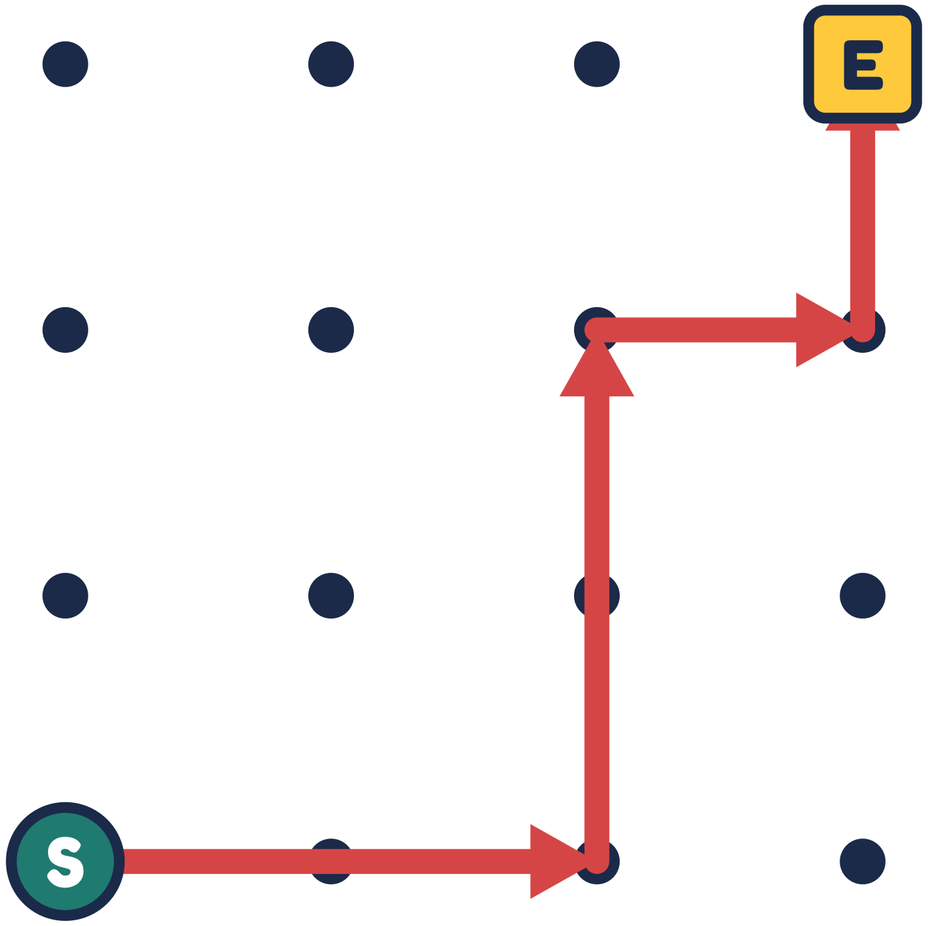
Put your button on the bottom left dot of a route grid.

1. Go 2 right.
2. Go 2 up.
3. Go 1 right.
4. Go 1 up.

Walk the button and draw the line as you go.

The line your button walked

12–23 MIN



Right, up, right, up. The button is on the top right dot.

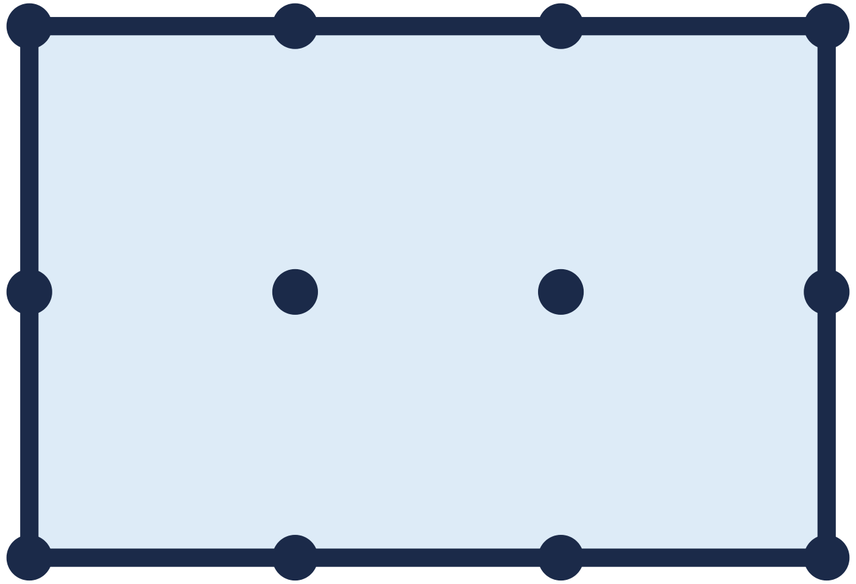
What did we notice?

12–23 MIN

- What did you count when you copied the house?
- What did you count when you walked the button?

Steps, not dots

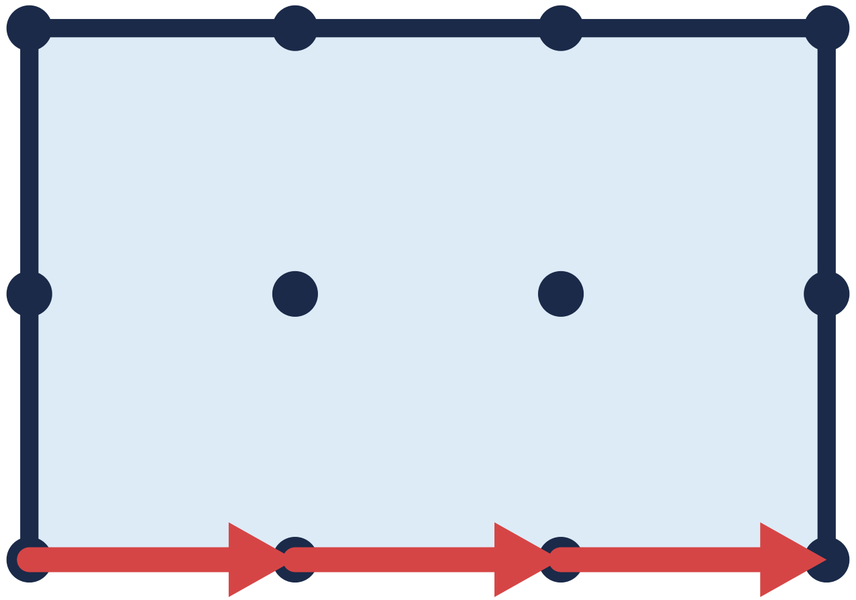
23–34 MIN



How many dots are on the bottom edge? How many steps long is it?

4 dots, 3 steps

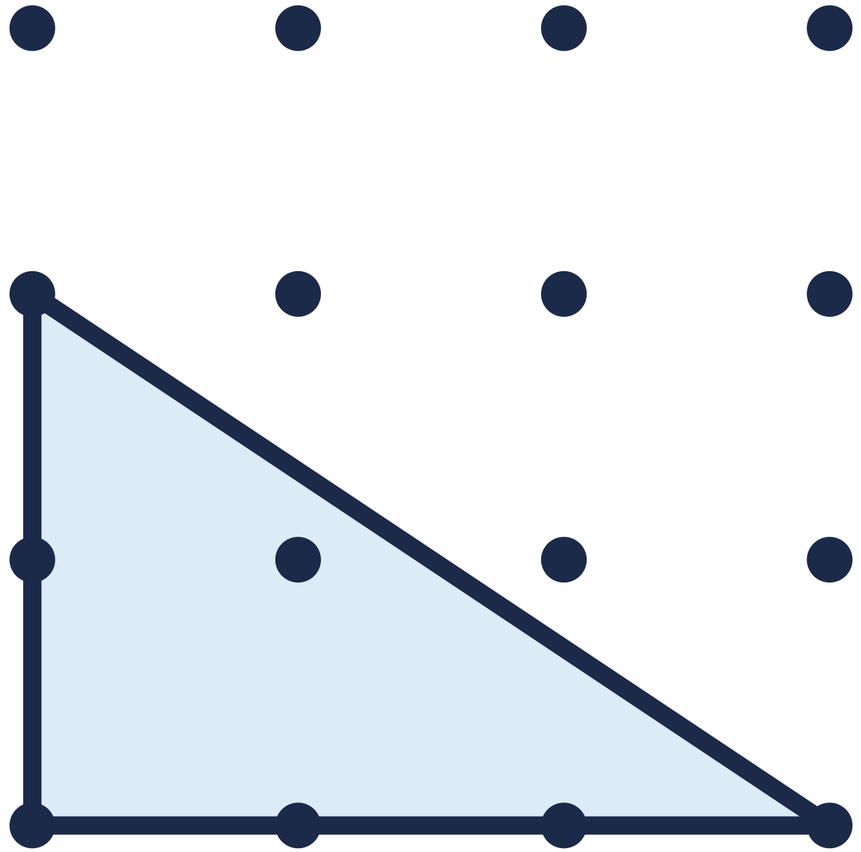
23–34 MIN



The bottom edge touches 4 dots. It is 3 steps long. We count steps, not dots.

Copy corner by corner

23–34 MIN

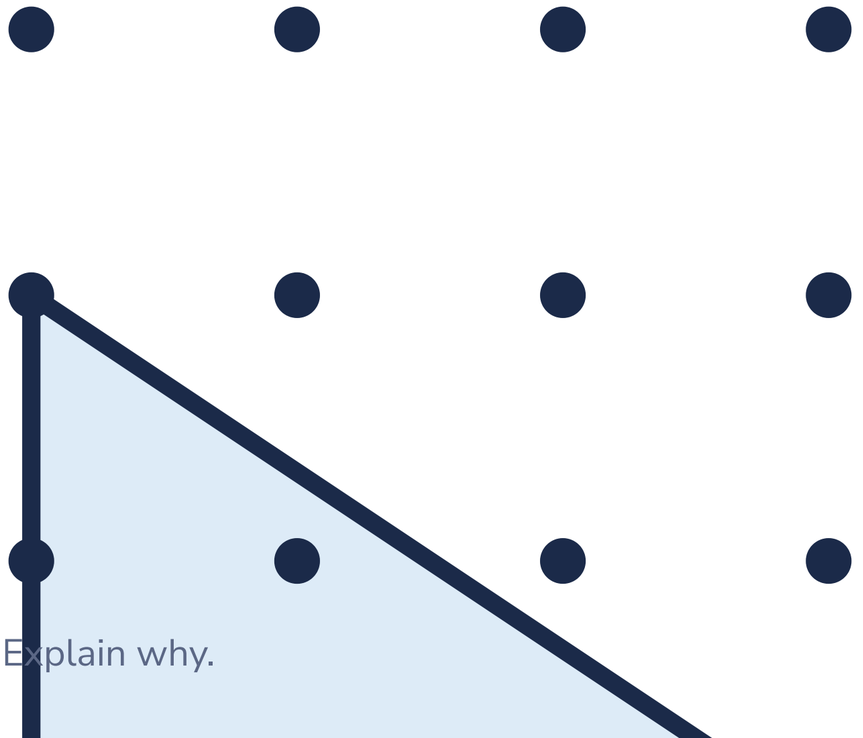


Corner A is the start, at the bottom left. How far right is corner B? How far up is corner C?

A, B and C

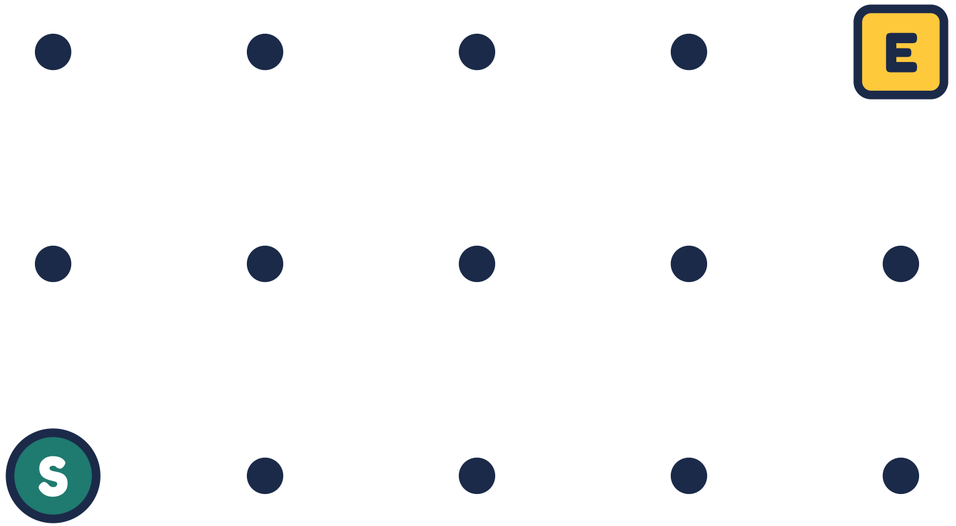
23–34 MIN

- A: the start.
- B: 3 right.
- C: 2 up.



Say where to go

23–34 MIN

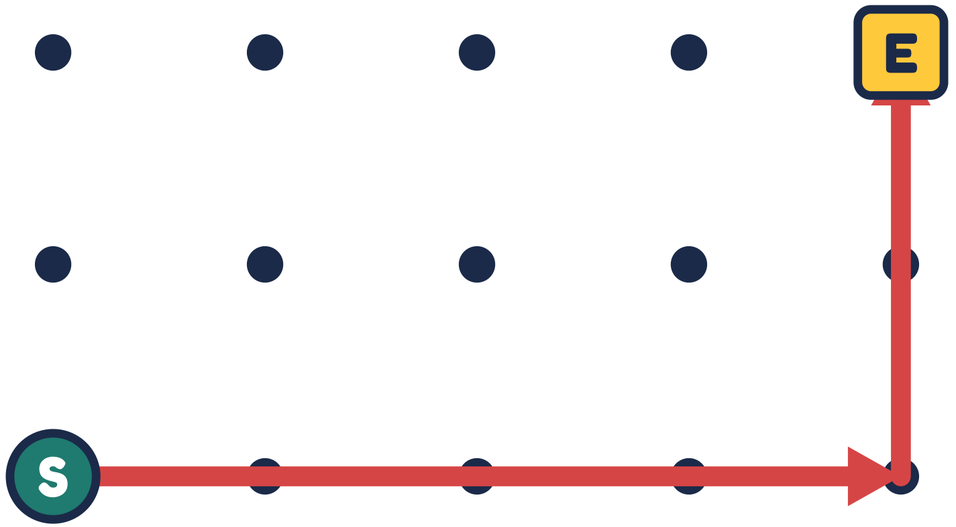


How do we get from S to E? ___ right, ___ up.

4 right, 2 up

23–34 MIN

4 right, 2 up. To come back: 4 left, 2 down.

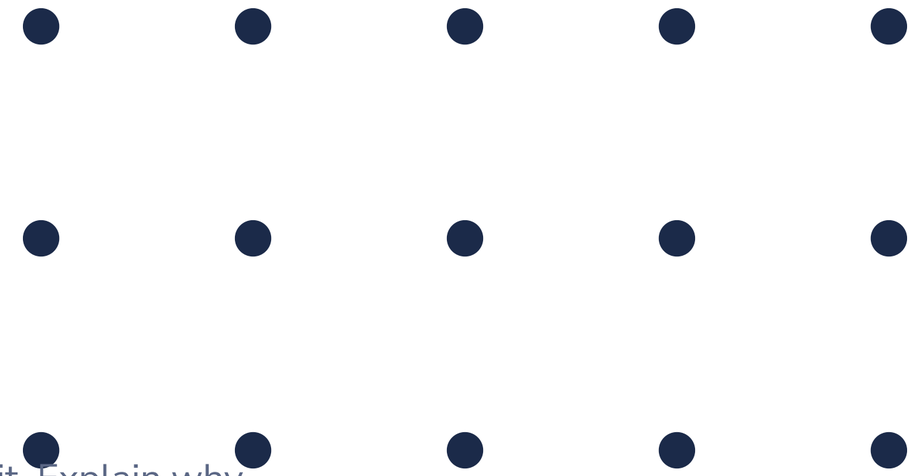


Follow the directions

23–34 MIN

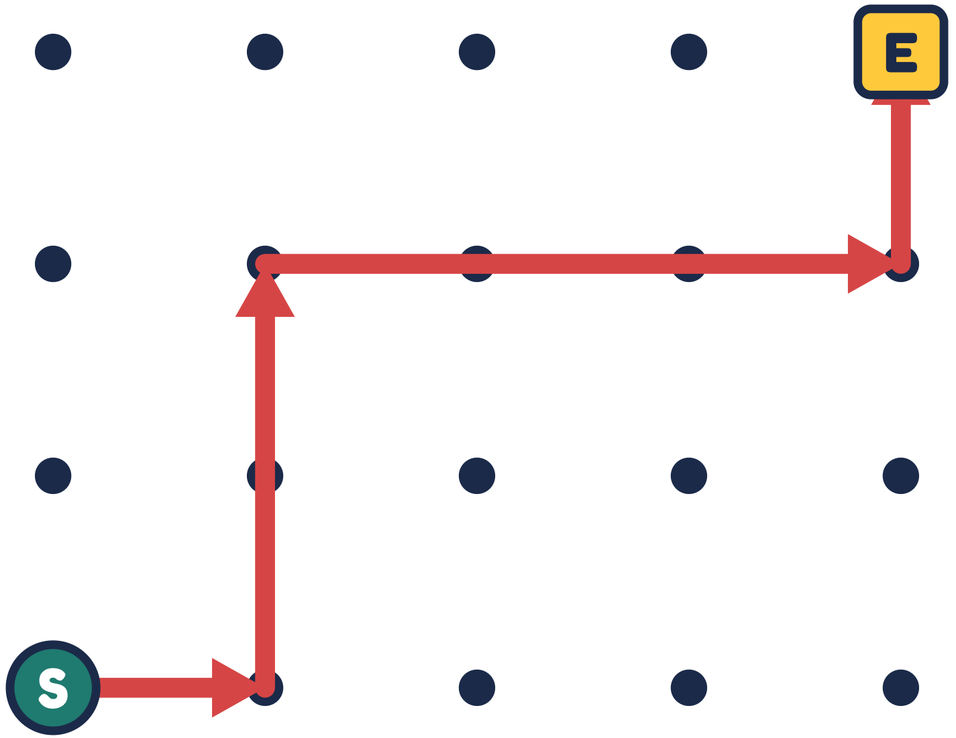
Start at S.

1. Go 1 right.
2. Go 2 up.
3. Go 3 right.
4. Go 1 up.



Where did we stop?

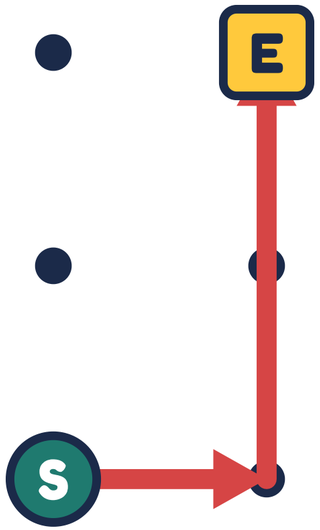
23–34 MIN



We stop at the top right dot. In all: 4 right and 3 up.

Different routes

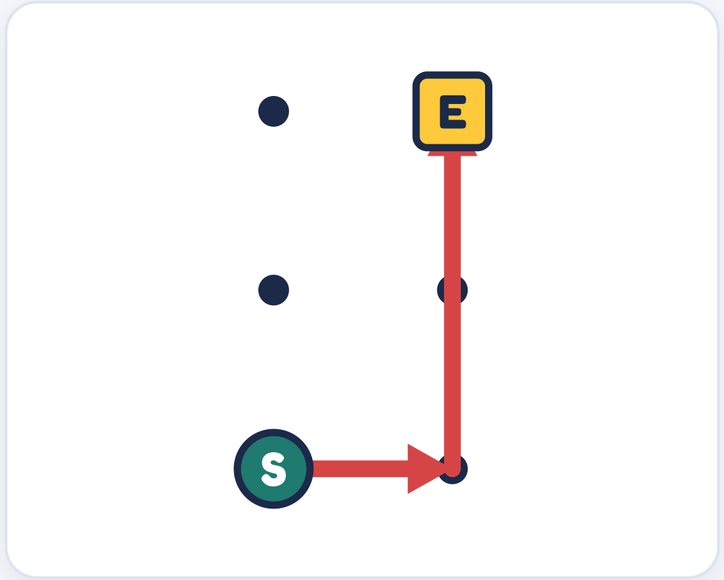
23–34 MIN



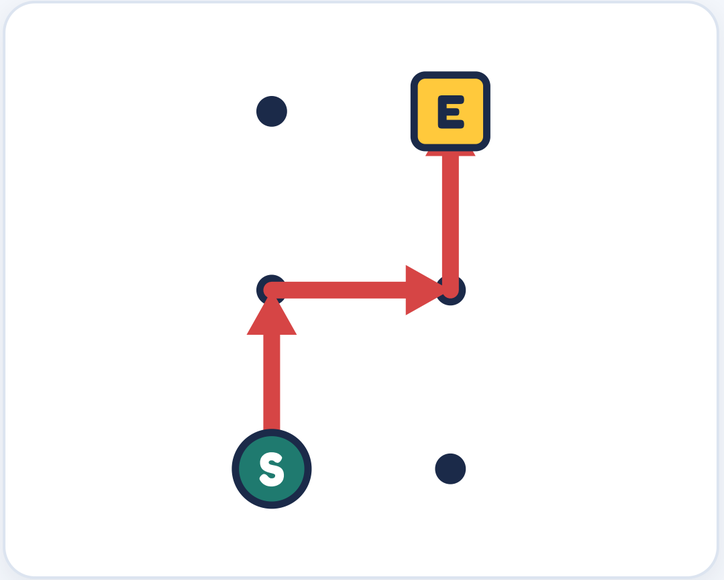
A route goes only right or up. This route is R, U, U. Find another route.

Three routes, no more

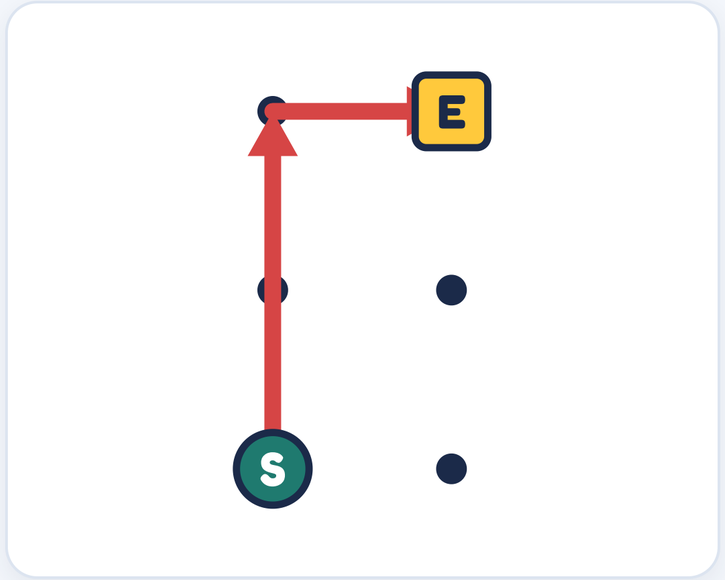
23–34 MIN



RUU



URU



UUR

The R can go first, second or third. That makes 3 routes.

34–38 MIN

Counter break

Stand up. Follow my directions with your body.

Take small steps right. Reach up high.

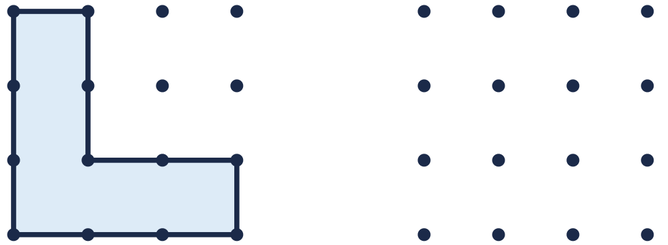
Sip some water.

Workbook page 1: Start here

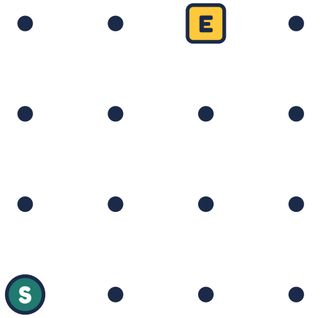
38–70 MIN

Questions 1 to 3. Take your time.

S1 Mei draws an L shape on a grid.



S2 A counter starts at S. It moves only right and up to reach E.



S3 Start at S. Go 4 steps right. Then go 1 step up.



38–70 MIN

Check page 1

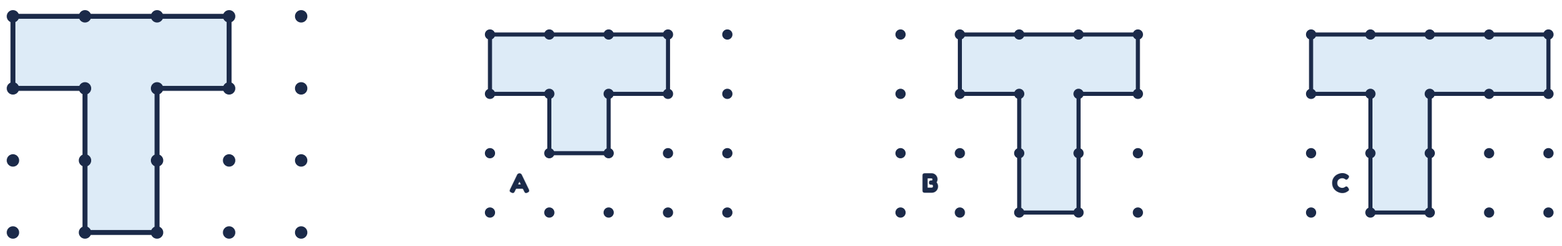
S1	A copy of the L shape; 3; 3
S2	2 right, 3 up; any path that goes only right and up
S3	A line 4 right and then 1 up; ring on the dot where it stops

Workbook page 2: Start here

38–70 MIN

Questions 4 to 6. Take your time.

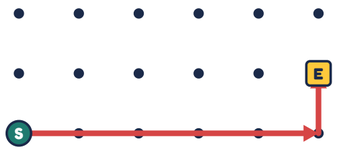
S4 Which picture is an exact copy of the T shape?



S5 Three corners of a rectangle are ringed. Join the dots to finish the rectangle.



S6 Look at the path from S to E. Tick True or False for each sentence.



38–70 MIN

Check page 2

S4	B
S5	3 right, 2 up
S6	True; False; False; True

38–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Count the routes, and a path that repeats
Go further (page 4)	Add at the corners, a corner the route must pass, your own figure
Investigation (page 5)	Same steps, different routes
Check	Finish and check your own work

Everyone may choose. Not finished? Pages 3 to 5 are for home too.

Talk and explain

70–76 MIN

TWO CHILDREN WALK FROM S

Ali goes 3 right, then 2 up.

Mei goes 2 up, then 3 right.

Do they stop at the same dot? Do they walk the same number of steps?

I think … because …

Back to the sneak-peek

70–76 MIN

Puzzle 1: 3 right, 1 up. Puzzle 2: 2 routes. Right then up, or up then right.

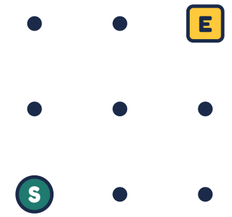
Exit ticket

76–82 MIN

On your own. No looking back.

E1 **A counter starts at S and moves only right and up to reach E.**

(a) The counter moves ☐ right and ☐ up.



E2 **A counter moves 2 right and 3 up. Then it moves 3 right and 1 up.**

(a) In all it moves ☐ right and ☐ up.

E3 **A figure is 5 steps across and 2 steps up. A child draws a copy that is 6 steps across and 2 steps up.**

(a) Is it an exact copy? Circle Yes or No.

(b) An exact copy is ☐ steps across.

Check the exit ticket

76–82 MIN

E1	2 right, 2 up
E2	5 right, 4 up
E3	No; 5

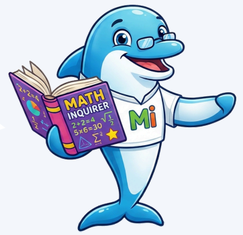
Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ counting steps, not dots ☐ copying exactly ☐ counting the routes ☐ something else
What helped me	☐ the grid page ☐ my button ☐ drawing the routes ☐ I was not stuck
Next time I will try	☐ counting steps ☐ saying where to go ☐ finding every route ☐ asking a question
	☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

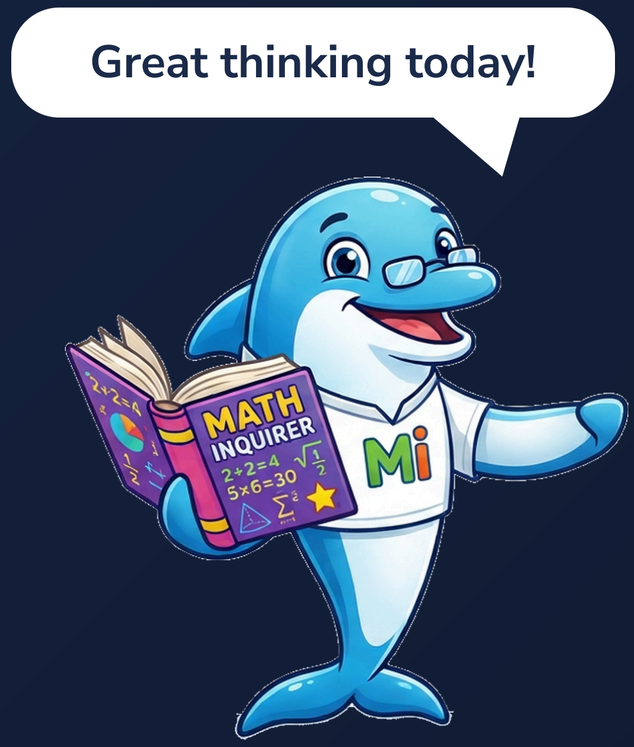
Well done, inquirers!

Show a grown-up how to walk from S to E on your grid page.

NEXT LESSON

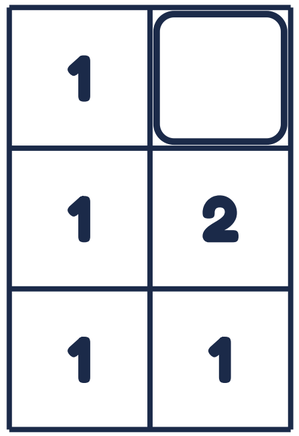
Cubes and Views

Have ready: 12 small blocks of one size (bricks or dice). The 4 paper squares from your book, cut out by a grown-up.



Go further: add at the corners

OPTIONAL



Each box says how many routes reach that corner. The empty box is the box on its left plus the box below it.

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	Does the order of the moves change where we stop?
Make a puzzle	Draw a grid puzzle for your grown-up.
Teach it back	Show a toy how to count steps, not dots.

Optional. Everyone may join.